

COMP0089 Reinforcement Learning

Exam Revision Checklist

Organized by Exam Topic Cluster
Based on 2021/22 – 2024/25 Past Papers

Auto-generated from Lecture Notes and Past Exams

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1 Exam Structure Overview

The exam consists of **4 mandatory questions** totaling 100 marks. The table below summarizes recurring topic clusters across all four years.

Topic Cluster	21/22	22/23	23/24	24/25	Marks	Lectures
Bandits (UCB, Thompson, Greedy)	Q1	Q1	Q1	Q1A	14–27	L2
UCB + Q-learning convergence	Q1d	Q1c	Q1c	–	7	L2, L6
Write update rules	Q2a	Q2a	Q2a	–	3–5	L3–L6
Grid MDP value/policy	Q2b	Q2b–c	Q2b–c	–	5–7	L3
Non-standard return $f(x) = x^\gamma$	Q2c	Q2d	Q2d	–	5–7	L3
Policy improvement theorem	Q2d	Q2e	Q2e	Q4d–e	4–7	L3, L4
Modified Bellman operator T	Q2e	Q2f	Q2f	Q4a–c	5–7	L4
TD(0) linear FA convergence	Q3b	Q3b	Q3b	Q1B	4–6	L5, L7
TD(λ) forward-view target	Q3a	Q3a	Q3a	–	2–3	L5
2-step TD convergence	Q3c	Q3c	–	–	4	L5, L6
Average-reward objective	Q4a	Q4a	–	–	2	L9
Optimal q^* simple MDP	Q4b	Q4b	–	–	2	L3, L4
Q-learning vs PG (partial obs.)	Q4b.ii	Q4b.ii	–	–	4–5	L6, L9
Gaussian policy gradient	Q4c	Q4c	–	–	4–8	L9
Non-standard policy param.	–	–	–	Q2	23	L9
SARSA vs Q-learning (maze)	–	–	–	Q3	24	L6
Reward shaping	–	–	–	Q3e	6	L6
Combining two policies	–	–	–	Q4d–e	12	L3, L4
Experience replay convergence	Q3b.iv	Q3b.iv	Q3b.iv	–	4–7	L7

Key observations:

- The modified Bellman operator question appeared in **all 4 years**—it is the single most predictable exam question.
- Policy improvement appeared every year, either as a standalone proof or embedded in a combining-policies question.
- The 2024/25 paper shifted significantly toward **conceptual reasoning** and away from template-based memorization. Only the modified Bellman operator was directly repeated.
- Lectures 8 (Planning/Dyna/MCTS), 10 (AVI/API bounds), and 11 (off-policy multi-step/V-trace) have **never** been directly examined.

2 Critical: Tested Every Year or in 2024/25

2.1 Modified Bellman Operator Analysis

CRITICAL

Exam history: Q2e (21/22, 7 marks), Q2f (22/23, 7 marks), Q2f (23/24, 7 marks), Q4a–c (24/25, 13 marks).

What the exam asks: State the operator T , determine whether it has a unique fixed point, whether it converges, compare with the standard Bellman optimality operator T^* , provide a counterexample in stochastic environments, and analyze the deterministic case.

The modified operator is defined as:

$$\forall s, a : \quad (Tq)(s, a) = r(s, a) + \gamma \max_b \mathbb{E}_{S_{t+1} \sim p(\cdot | s, a)} [q(S_{t+1}, b)] \quad (1)$$

where $r(s, a) = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$.

Compare with the standard Bellman optimality operator:

$$\forall s, a : \quad (T^*q)(s, a) = r(s, a) + \gamma \mathbb{E}_{S_{t+1} \sim p(\cdot | s, a)} \left[\max_b q(S_{t+1}, b) \right] \quad (2)$$

Key results:

1. T is a γ -contraction under $\|\cdot\|_\infty$ (proven identically to T^* , since $|\max_b f(b) - \max_b g(b)| \leq \max_b |f(b) - g(b)|$).
2. By the Banach Fixed Point Theorem, T has a **unique** fixed point $\hat{q}_T$ and synchronous iteration converges to it.
3. In general, $\hat{q}_T \neq q^*$ because $\max \mathbb{E}[\cdot] \leq \mathbb{E}[\max(\cdot)]$ (Jensen's inequality for the convex max function).
4. In **deterministic** environments, $\mathbb{E}[\cdot]$ is a point evaluation, so $\max \mathbb{E}[\cdot] = \mathbb{E}[\max(\cdot)]$ and thus $T = T^*$ and $\hat{q}_T = q^*$.
5. In stochastic environments, the fixed point satisfies $\hat{q}_T(s, a) \leq q^*(s, a)$ for all (s, a) , because T uses a “pessimistic” backup.
6. The greedy policy w.r.t. $\hat{q}_T$ is not guaranteed to be optimal (and not guaranteed to improve over the policy induced by any arbitrary q).

Self-check: Can I state this from memory? ☐ Yes ☐ Partially ☐ No

Can I reproduce the contraction proof? ☐ Yes ☐ Partially ☐ No

2.2 Policy Improvement Theorem

CRITICAL

Exam history: Q2d (21/22, 6 marks), Q2e (22/23, 6 marks), Q2e (23/24, 6 marks), Q4d–e (24/25, 12 marks).

What the exam asks: Prove that the greedy policy $\pi'(a|s) = \arg \max_a q_\pi(s, a)$ satisfies $v_{\pi'}(s) \geq v_\pi(s)$ for all s . State when equality holds. In 24/25: apply to combining two policies.

Theorem 1 (Policy Improvement Theorem — Lecture 3). *Let π and π' be any pair of deterministic policies such that:*

$$q_\pi(s, \pi'(s)) \geq v_\pi(s), \quad \forall s \in \mathcal{S} \quad (3)$$

Then π' is as good as or better than π :

$$v_{\pi'}(s) \geq v_\pi(s), \quad \forall s \in \mathcal{S} \quad (4)$$

Proof sketch:

$$v_\pi(s) \leq q_\pi(s, \pi'(s)) \quad (5)$$

$$= \mathbb{E}_{\pi'}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s] \quad (6)$$

$$\leq \mathbb{E}_{\pi'}[R_{t+1} + \gamma q_\pi(S_{t+1}, \pi'(S_{t+1})) \mid S_t = s] \quad (7)$$

$$= \mathbb{E}_{\pi'}[R_{t+1} + \gamma R_{t+2} + \gamma^2 v_\pi(S_{t+2}) \mid S_t = s] \quad (8)$$

$$\leq \dots \leq \mathbb{E}_{\pi'}[G_t \mid S_t = s] = v_{\pi'}(s) \quad (9)$$

Key step: repeatedly apply the greedy inequality $v_\pi(s) \leq q_\pi(s, \pi'(s))$ to unroll.

Equality holds when π is already optimal: $v_\pi(s) = v^*(s)$ for all s .

2024/25 extension (combining two policies): Given π_1, π_2 with non-ordered values, define $\pi(a|s) = \arg \max_a \max(q_{\pi_1}(s, a), q_{\pi_2}(s, a))$. This is a greedy policy w.r.t. $\hat{q}(s, a) = \max(q_{\pi_1}(s, a), q_{\pi_2}(s, a))$.

Since $\hat{q}(s, \pi(s)) \geq q_{\pi_i}(s, \pi_i(s)) = v_{\pi_i}(s)$ for each i , the greedy policy π improves on **both** π_1 and π_2 .

To construct a policy improving on both: use $\pi'(a|s) = \arg \max_a \max(q_{\pi_1}(s, a), q_{\pi_2}(s, a))$.

Self-check: Can I state this from memory? ☐ Yes ☐ Partially ☐ No

Can I reproduce the proof? ☐ Yes ☐ Partially ☐ No

2.3 Banach Fixed Point Theorem and Contraction Mappings

CRITICAL

Exam history: Underlies the modified Bellman operator proof every year. Also needed for Q-learning convergence arguments.

What the exam asks: Apply Banach FPT to prove convergence and uniqueness of fixed points for Bellman-type operators.

Definition 2 (Contraction Mapping — Lecture 4). Let $\mathcal{X}$ be a vector space equipped with a norm $\|\cdot\|$. A mapping $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{X}$ is an α -contraction mapping if for any $x_1, x_2 \in \mathcal{X}$, there exists $\alpha \in [0, 1)$ such that:

$$\|\mathcal{T}x_1 - \mathcal{T}x_2\| \leq \alpha \|x_1 - x_2\| \quad (10)$$

Theorem 3 (Banach Fixed Point Theorem — Lecture 4). Let $\mathcal{X}$ be a complete normed vector space equipped with a norm $\|\cdot\|$ and $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{X}$ be a γ -contraction mapping, then:

1. $\mathcal{T}$ has a unique fixed point $x^* \in \mathcal{X}$: $\exists! x^* \in \mathcal{X}$ such that $\mathcal{T}x^* = x^*$.
2. Iterating $\mathcal{T}$ from any initial point x_0 converges: $\lim_{n \rightarrow \infty} \mathcal{T}^n x_0 = x^*$.
3. The convergence rate is geometric: $\|\mathcal{T}^n x_0 - x^*\| \leq \gamma^n \|x_0 - x^*\|$.

Critical application: Both T^π and T^* (and the modified T) are γ -contractions under $\|\cdot\|_\infty$, so Banach FPT guarantees unique fixed points v_π , q^* , and $\hat{q}_T$ respectively.

Important distinction: Banach FPT justifies the *deterministic* fixed point. Stochastic convergence (e.g., Q-learning, TD(0)) requires the *stochastic approximation theorem* (Robbins–Monro conditions) separately.

Self-check: Can I state this from memory? ☐ Yes ☐ Partially ☐ No

Can I reproduce the contraction proof for T^* ? ☐ Yes ☐ Partially ☐ No

2.4 Bellman Operators: Definitions and Contraction Proofs

CRITICAL

Exam history: Foundation for Q2 every year. Update rules, value computations, and operator analysis all rely on these.

Definition 4 (Bellman Expectation Operator T^π — Lecture 4). For any policy π , the Bellman Expectation operator $T_V^\pi : \mathcal{V} \rightarrow \mathcal{V}$ is defined point-wise as:

$$(T_V^\pi f)(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) f(s') \right] \quad (11)$$

Unique fixed point: v_π . Contraction factor: γ .

Definition 5 (Bellman Optimality Operator T^* — Lecture 4).

$$(T_V^* f)(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) f(s') \right] \quad (12)$$

Unique fixed point: v^* . Contraction factor: γ .

Definition 6 (Bellman Optimality Operator for Q — Lecture 4).

$$(T_Q^* f)(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \max_{a'} f(s', a') \quad (13)$$

Unique fixed point: q^* . Contraction factor: γ .

Contraction proof for T_Q^* (key steps):

$$\|T^* q_1 - T^* q_2\|_\infty = \max_{s,a} \left| \gamma \sum_{s'} p(s'|s, a) \left[\max_{a'} q_1(s', a') - \max_{a'} q_2(s', a') \right] \right| \quad (14)$$

$$\leq \gamma \max_{s,a} \sum_{s'} p(s'|s, a) \max_{a'} |q_1(s', a') - q_2(s', a')| \quad (15)$$

$$\leq \gamma \|q_1 - q_2\|_\infty \quad (16)$$

The critical inequality: $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$.

Self-check: Can I state all three operators from memory? ☐ Yes ☐ Partially ☐ No

Can I prove T^* is a γ -contraction? ☐ Yes ☐ Partially ☐ No

2.5 Bandits: UCB, Thompson Sampling, Greedy

CRITICAL

Exam history: Q1 every year (14–27 marks total). UCB calculation with specific data, Thompson Sampling posterior computation, greedy policy evaluation. In 24/25: three-armed bandit with negative rewards, expected UCB probabilities.

UCB Algorithm (Lecture 2):

$$A_t = \arg \max_{a \in \mathcal{A}} \left[Q_t(a) + c \sqrt{\frac{\log t}{N_t(a)}} \right] \quad (17)$$

where $Q_t(a)$ is the sample mean, $N_t(a)$ is the count, and c is the exploration parameter.

Thompson Sampling (Lecture 2): With Bernoulli rewards and a uniform Beta(1,1) prior: after observing k successes and ℓ failures for action a , the posterior is Beta($1+k, 1+\ell$).

Thompson Sampling: sample $\tilde{Q}(a) \sim \text{Beta}(\alpha_a, \beta_a)$ for each a , select $A_t = \arg \max_a \tilde{Q}(a)$.

Key computation: $\Pr(A_t = a) = \Pr(\tilde{Q}(a) > \tilde{Q}(b))$ requires integrating $\int_0^1 p_a(x) \Pr(\tilde{Q}(b) < x) dx$ where $\Pr(\tilde{Q}(b) < x)$ is the CDF of the Beta distribution.

Warning: Swapping α/β in the Beta posterior is **not** a convention choice—it changes the argmax direction and produces wrong answers.

Self-check: Can I compute UCB action selection for specific data? ☐ Yes ☐ Partially ☐ No

Can I compute Thompson Sampling probabilities with Beta posteriors? ☐ Yes ☐ Partially ☐ No

2.6 TD(0) with Linear Function Approximation: Convergence Analysis

CRITICAL

Exam history: Q3b every year (4–6 marks each year), Q1B in 24/25 (13 marks). Two-state MRP/MDP, scalar weight, compute $\mathbb{E}[w_t]$, analyze convergence/divergence, on-policy vs off-policy (β parameter), experience replay.

TD(0) update with linear FA (Lecture 5/7): For $v_{\mathbf{w}}(s) = \mathbf{w}^\top \mathbf{x}(s)$,

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha \left(R_{t+1} + \gamma \mathbf{w}_t^\top \mathbf{x}_{t+1} - \mathbf{w}_t^\top \mathbf{x}_t \right) \mathbf{x}_t \quad (18)$$

Expected update (scalar case): With $v_w(s) = w \cdot \phi(s)$,

$$\mathbb{E}[w_{t+1}] = \mathbb{E}[w_t] + \alpha \mathbb{E}[(R_{t+1} + \gamma w_t \phi(S_{t+1}) - w_t \phi(S_t)) \phi(S_t)] \quad (19)$$

TD Fixed Point (Lecture 7):

$$\mathbf{w}_{\text{TD}} = \mathbb{E}[\mathbf{x}_t(\mathbf{x}_t - \gamma \mathbf{x}_{t+1})^\top]^{-1} \mathbb{E}[R_{t+1} \mathbf{x}_t] \quad (20)$$

Convergence condition: Requires the matrix $\mathbf{A} = \mathbb{E}[\mathbf{x}_t(\mathbf{x}_t - \gamma \mathbf{x}_{t+1})^\top]$ to have eigenvalues with **negative real parts** (for the continuous-time ODE analogy) or equivalently $\rho(\mathbf{I} + \alpha \mathbf{A}) < 1$ for the chosen step size α .

Off-policy sampling (β parameter): When sampling states from a distribution different from the on-policy stationary distribution, the matrix $\mathbf{A}$ changes. Convergence depends on whether the resulting $\mathbf{A}$ still has the correct spectral properties. This is the **deadly triad**: off-policy + function approximation + bootstrapping can diverge.

Experience replay: Converges to the **maximum-likelihood MDP solution** if the replay buffer contains all transitions. With a finite sample of n transitions, convergence depends on whether the empirical sampling distribution matches the on-policy distribution. If the buffer lacks transitions from some states, convergence may fail.

Self-check: Can I compute $\mathbb{E}[w_1]$ for a concrete two-state problem? ☐ Yes ☐ Partially ☐ No

Can I analyze convergence as a function of β ? ☐ Yes ☐ Partially ☐ No

2.7 Policy Gradient: Score Function Trick and PGT

CRITICAL

Exam history: Q4 in 21/22–23/24, Q2 in 24/25 (23 marks). Score function derivation, numerical update computation, on-policy action selection benefit, Gaussian policy gradient.

Lemma 7 (Score Function Trick — Lecture 9). *Let d be a state distribution independent of θ . Then*

$$\nabla_{\theta} \mathbb{E}_{\pi_{\theta}}[R(S, A)] = \mathbb{E}_{\pi_{\theta}}[R(S, A) \nabla_{\theta} \log \pi_{\theta}(A | S)]. \quad (21)$$

Derivation:

$$\nabla_{\theta} \sum_a \pi_{\theta}(a|s) r(s, a) = \sum_a r(s, a) \nabla_{\theta} \pi_{\theta}(a|s) \quad (22)$$

$$= \sum_a \pi_{\theta}(a|s) r(s, a) \frac{\nabla_{\theta} \pi_{\theta}(a|s)}{\pi_{\theta}(a|s)} \quad (23)$$

$$= \mathbb{E}_{\pi} [R(S, A) \nabla_{\theta} \log \pi_{\theta}(A | S)] \quad (24)$$

Theorem 8 (Policy Gradient Theorem — Episodic (Lecture 9)).

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^T \gamma^t q_{\pi_{\theta}}(S_t, A_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t) \mid S_0 \sim d_0 \right] \quad (25)$$

Theorem 9 (Policy Gradient Theorem — Average Reward (Lecture 9)).

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi}[q_{\pi_{\theta}}(S_t, A_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t)] \quad (26)$$

where the differential action-value: $q_{\pi}(s, a) = \mathbb{E}_{\pi}[R_{t+1} - \rho + q_{\pi}(S_{t+1}, A_{t+1}) | S_t = s, A_t = a]$, $\rho = \mathbb{E}_{\pi}[R_{t+1}]$.

Proposition 10 (Baseline Does Not Change Expected Gradient — Lecture 9). For any function $b(S_t)$ independent of action A_t :

$$\mathbb{E}_{\pi}[b(S_t) \nabla_{\theta} \log \pi_{\theta}(A_t | S_t)] = 0. \quad (27)$$

Proof: $\sum_a \nabla_{\theta} \pi_{\theta}(a|s) = \nabla_{\theta} \sum_a \pi_{\theta}(a|s) = \nabla_{\theta} 1 = 0$.

Past rewards vanish: For $k < t$, R_{k+1} is independent of A_t conditional on S_t . By the tower rule and the zero-mean property of the score function, $\mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(A_t | S_t) \cdot R_{k+1} | S_t] = \mathbb{E}[R_{k+1} | S_t] \cdot \mathbb{E}[\nabla_{\theta} \log \pi_{\theta}(A_t | S_t) | S_t] = 0$.

On-policy action selection (24/25, Q2d): Using $A_t \sim \pi_{\theta}$ gives an **unbiased** gradient estimator because $\mathbb{E}_{\pi}[R \nabla \log \pi(A|S)] = \nabla_{\theta} J(\theta)$ only holds when actions are sampled from π .

Gaussian policy score function: For $\pi(a|s) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(a-\mu)^2}{2\sigma^2}\right)$:

$$\nabla_{\mu} \log \pi(a|s) = \frac{a - \mu}{\sigma^2}, \quad \nabla_{\sigma} \log \pi(a|s) = \frac{(a - \mu)^2 - \sigma^2}{\sigma^3} \quad (28)$$

Gaussian PG update: $\mu' = \mu + \alpha G \cdot \frac{A - \mu}{\sigma^2}$.

Entropy increases when: $|A - \mu| > \sigma$ (for $G > 0$), because the variance update pushes σ larger.

Self-check: Can I derive the score function trick from scratch? ☐ Yes ☐ Partially ☐ No

Can I compute Gaussian policy updates numerically? ☐ Yes ☐ Partially ☐ No

2.8 SARSA vs Q-learning: On-Policy vs Off-Policy

CRITICAL

Exam history: Q3 in 24/25 (24 marks). SARSA evaluates the **actual** ε -greedy policy; Q-learning learns q^* regardless of the behavior policy. Iterated greedy improvement.

SARSA update (Lecture 6):

$$q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha (R_{t+1} + \gamma q_t(S_{t+1}, A_{t+1}) - q_t(S_t, A_t)) \quad (29)$$

where $A_{t+1} \sim \pi(\cdot | S_{t+1})$ is the **actually taken** next action.

Q-learning update (Lecture 6):

$$q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha \left(R_{t+1} + \gamma \max_{a'} q_t(S_{t+1}, a') - q_t(S_t, A_t) \right) \quad (30)$$

Key distinction:

- SARSA: on-policy, evaluates q_{π} for the current (e.g., ε -greedy) policy. Its converged values reflect the *risk* of the behavior policy.
- Q-learning: off-policy, converges to q^* regardless of exploration. The greedy policy from q^* is always optimal.
- Both execute ε -greedy actions in the environment; the difference is only in the **update target**.

Cliff-walking / maze insight: SARSA's cliff-edge avoidance is **not** a local optimum—it correctly evaluates the ε -greedy policy under which the cliff-edge path is genuinely risky.

Iterated SARSA improvement: $\pi \rightarrow q_{\pi} \rightarrow \pi' \rightarrow q_{\pi'} \rightarrow \dots$ converges, but each intermediate greedy policy may not go to the globally optimal path.

Iterated Q-learning: After convergence to q^* , the greedy policy is immediately optimal; further iterations do not change it.

Self-check: Can I explain the behavioral difference in a maze? ☐ Yes ☐ Partially ☐ No

2.9 Reward Shaping

CRITICAL

Exam history: Q3e (24/25, 6 marks). Manhattan-distance-based reward shaping. Does the reshaped problem's Q-learning solution match intended behavior?

Potential-based reward shaping (Ng et al., 1999): Define $F(s, s') = \gamma\Phi(s') - \Phi(s)$ for some potential function Φ . The shaped reward $r'(s, a, s') = r(s, a, s') + F(s, s')$ preserves the optimal policy.

Exam application: When shaping is **not** potential-based (e.g., Manhattan-distance reward $20 - m$ at each step), Q-learning on the reshaped MDP converges to the optimal policy *of the reshaped MDP*, which may differ from the intended behavior. One must check whether the reshaped problem's greedy policy actually takes the desired path.

Self-check: Can I analyze whether a specific shaping changes the optimal policy? ☐ Yes ☐ Partially ☐ No

3 High Priority: Tested 1–2 Years or Key Prerequisite

3.1 Value Function Definitions and Bellman Equations

HIGH

Exam history: Q2a every year (update rules), Q2b (grid MDP calculations). Prerequisite for nearly everything.

Definition 11 (State-Value Function — Lecture 3). $v_\pi(s) = \mathbb{E}_\pi [\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s]$

Definition 12 (Action-Value Function — Lecture 3). $q_\pi(s, a) = \mathbb{E}_\pi [\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a]$

Bellman Expectation Equations (Lecture 3):

$$v_\pi(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) v_\pi(s') \right] \quad (31)$$

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \sum_{a'} \pi(a'|s') q_\pi(s', a') \quad (32)$$

Bellman Optimality Equations (Lecture 3):

$$v^*(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|a, s) v^*(s') \right] \quad (33)$$

$$q^*(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|a, s) \max_{a'} q^*(s', a') \quad (34)$$

Self-check: Can I write all four equations from memory? ☐ Yes ☐ Partially ☐ No

3.2 TD(λ) Forward-View Target

HIGH

Exam history: Q3a in 21/22–23/24 (2–3 marks). Write the λ -return in recursive and explicit summation form.

λ -Return (Lecture 5):

$$G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)} \quad (35)$$

Recursive form:

$$G_t^\lambda = R_{t+1} + \gamma \left[(1 - \lambda) v(S_{t+1}) + \lambda G_{t+1}^\lambda \right] \quad (36)$$

Explicit summation form:

$$G_t^\lambda = \sum_{n=0}^{\infty} (\gamma \lambda)^n [R_{t+n+1} + \gamma(1 - \lambda) v(S_{t+n+1})] \quad (37)$$

(with the convention that $v(S_T) = 0$ at terminal states, so the sum terminates).

Special cases: $\lambda = 0$ recovers TD(0), $\lambda = 1$ recovers MC.

Self-check: Can I write both forms from memory? ☐ Yes ☐ Partially ☐ No

3.3 UCB + Q-learning Convergence in Tabular MDP

HIGH

Exam history: Q1d (21/22, 7 marks), Q1c (22/23, 7 marks), Q1c (23/24, 7 marks). Not in 24/25.

Argument: In an ergodic tabular MDP, using UCB locally in each state to select actions while updating q values with Q-learning:

1. UCB guarantees that every action in every state is selected infinitely often (exploration guarantee from the $\sqrt{\log t / N_t(a)}$ term growing without bound for unvisited actions).
2. Ergodicity ensures every state is visited infinitely often.
3. Therefore every (s, a) pair is visited infinitely often.
4. Q-learning convergence theorem: given all (s, a) visited infinitely often and appropriate step sizes ($\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$), Q-learning converges to q^* .

Self-check: Can I write this proof concisely? ☐ Yes ☐ Partially ☐ No

3.4 Non-Standard Return / Discounting

HIGH

Exam history: Q2c/d in 21/22–23/24 (5–7 marks). Prove no standard γ makes going to +2 optimal; check if $f(x) = x^\gamma$ fixes it.

Template: For the grid MDP with rewards +1 (1 step), +2 (2 steps), + X (3 steps):

- Going right to +2: $v_{\text{up}} = \gamma^2 \cdot 2 = 2\gamma^2$ (2 steps).
- Going down to + X : $v_{\text{down}} = \gamma^3 \cdot X$ (3 steps, varies by year).
- Going right to +1: $v_{\text{right}} = \gamma \cdot 1 = \gamma$ (1 step).

To prove “no γ makes +2 optimal”: show $2\gamma^2 \leq \max(\gamma, X\gamma^3)$ for all $\gamma \in [0, 1]$.

Power return $f(x) = x^\gamma$: The return becomes $G = (R_1 + (R_2 + \dots)^\gamma)^\gamma$, which is nonlinear and can create different orderings.

Self-check: Can I set up the inequalities for a given X ? ☐ Yes ☐ Partially ☐ No

3.5 Average-Reward Policy Objective

HIGH

Exam history: Q4a in 21/22–22/23 (2 marks).

Definition 13 (Average-Reward Objective — Lecture 9).

$$J(\theta) = \mathbb{E}_{\pi_\theta}[R_{t+1}] = \sum_s d_\pi(s) \sum_a \pi_\theta(a|s) r(s, a) \quad (38)$$

where $d_\pi(s) = \lim_{t \rightarrow \infty} \Pr(S_t = s | \pi)$ is the stationary distribution under π .

Self-check: Can I state this from memory? ☐ Yes ☐ Partially ☐ No

3.6 Q-Learning vs Policy Gradient under Partial Observability

HIGH

Exam history: Q4b.ii in 21/22–22/23 (4–5 marks).

When states are aliased (both s_1 and s_2 map to the same observation s):

- **Q-learning** with ε -greedy: converges to a **deterministic** policy (greedy w.r.t. q). With state aliasing, the optimal deterministic policy may be suboptimal.
- **Policy gradient**: can learn a **stochastic** policy, mixing actions optimally across aliased states. Therefore, policy gradient is preferred under partial observability.

Self-check: Can I explain why stochastic policies help with state aliasing? ☐ Yes ☐ Partially ☐ No

3.7 TD Value Error Bound

HIGH

Prerequisite for understanding TD fixed point quality.

Theorem 14 (TD Value Error Bound — Lecture 7). *The TD fixed point $\mathbf{w}_{\text{TD}}$ satisfies:*

$$\overline{\text{VE}}(\mathbf{w}_{\text{TD}}) \leq \frac{1}{1-\gamma} \overline{\text{VE}}(\mathbf{w}_{\text{MC}}) = \frac{1}{1-\gamma} \min_{\mathbf{w}} \overline{\text{VE}}(\mathbf{w}) \quad (39)$$

where $\overline{\text{VE}}(\mathbf{w}) = \mathbb{E}_{S \sim d_{\pi}} [(v_{\pi}(S) - v_{\mathbf{w}}(S))^2]$.

MC finds the best linear approximation (projection onto feature space). TD's fixed point is at most $1/(1-\gamma)$ times worse.

Self-check: Can I state this bound? ☐ Yes ☐ Partially ☐ No

3.8 2-Step TD Target Convergence

HIGH

Exam history: Q3c in 21/22–22/23 (4 marks).

The target $\delta_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 \max_a q(S_{t+2}, a) - q(S_t, A_t)$:

This is a 2-step Q-learning target. Under tabular setting with appropriate step sizes and full coverage of (s, a) pairs, it converges to q^* . The proof follows the same structure as standard Q-learning convergence: the 2-step Bellman optimality operator $T^{(2)*}$ is a γ^2 -contraction, and $(T^{(2)*})$ has the same fixed point q^* as T^* .

Self-check: Can I state what this converges to and why? ☐ Yes ☐ Partially ☐ No

3.9 Gaussian Policy Gradient Details

HIGH

Exam history: Q4c in 21/22–22/23 (4–8 marks). Compute μ' after one update; when does entropy increase?

For $\pi(a|s) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(a-\mu)^2}{2\sigma^2}\right)$:

$$\mu' = \mu + \alpha G \cdot \frac{A - \mu}{\sigma^2} \quad (40)$$

$$\sigma' = \sigma + \alpha G \cdot \frac{(A - \mu)^2 - \sigma^2}{\sigma^3} \quad (41)$$

Entropy of Gaussian: $H = \frac{1}{2} \log(2\pi e \sigma^2)$. Entropy increases iff $\sigma' > \sigma$, which (for $G > 0$) requires $(A - \mu)^2 > \sigma^2$, i.e., $|A - \mu| > \sigma$.

Self-check: Can I derive these updates? ☐ Yes ☐ Partially ☐ No

3.10 Optimal q^* Computation for Simple MDPs

HIGH

Exam history: Q4b.i in 21/22–22/23 (2 marks). Two-state MDP with left/right/down.

For the standard exam MDP with s_1, s_2 , three actions (left, right, down), discount $\gamma = 0.9$: Set up $q^*(s, a)$ using the Bellman optimality equation and solve the resulting system. For “down” (terminal): $q^*(s_1, \text{down}) = -1$, $q^*(s_2, \text{down}) = +1$. For left/right: solve the linear system from the recursive Bellman equations.

Self-check: Can I set up and solve such systems? ☐ Yes ☐ Partially ☐ No

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4 Medium Priority: In Examinable Lectures, Never Directly Tested

MEDIUM

These topics appear in Lectures 1–9 but have never been directly examined. They may appear as novel questions in future exams.

Softmax policy score function (Lecture 2/9): $\frac{\partial \log \pi(A_t)}{\partial H(a)} = \mathcal{I}(a = A_t) - \pi(a)$.

Deterministic Policy Gradient (Lecture 9): $\nabla_{\theta} J(\theta) = \mathbb{E}_{S \sim d_{\mu}} [\nabla_{\theta} \mu_{\theta}(S) \nabla_a q_{\mu}(S, a)|_{a=\mu_{\theta}(S)}]$.

LSTD Closed-Form (Lecture 7): $\mathbf{w}_{\text{TD}} = \mathbf{A}^{-1} \mathbf{b}$ where $\mathbf{A} = \sum_t \mathbf{x}_t (\mathbf{x}_t - \gamma \mathbf{x}_{t+1})^{\top}$, $\mathbf{b} = \sum_t R_{t+1} \mathbf{x}_t$.

Actor-Critic (Lecture 9): Uses a learned value function $v_{\mathbf{w}}(s)$ as baseline and/or for bootstrapping the TD error $\delta_t = R_{t+1} + \gamma v_{\mathbf{w}}(S_{t+1}) - v_{\mathbf{w}}(S_t)$.

Double Q-learning (Lecture 6): Addresses overestimation bias in Q-learning by maintaining two independent value functions.

Expected SARSA (Lecture 6): Target $R_{t+1} + \gamma \sum_{a'} \pi(a'|S_{t+1}) q(S_{t+1}, a')$. On-policy version uses current policy; off-policy version uses target policy π_{target} .

Deadly Triad (Lecture 7/11): The combination of bootstrapping + function approximation + off-policy learning can cause divergence.

Lai-Robbins lower bound (Lecture 2): $\liminf_{t \rightarrow \infty} \frac{\mathbb{E}[L_t]}{\log t} \geq \sum_{a: \Delta_a > 0} \frac{\Delta_a}{\text{KL}(\mathcal{R}_a \| \mathcal{R}_{a^*})}$.

5 Low Priority: Never Examined (Lectures 8, 10, 11)

SKIP

These topics have never appeared on any past exam paper. Listed here for completeness only.

Lecture 8 — Planning and Models: Dyna-Q, model-based RL, Monte Carlo Tree Search (MCTS), forward search, simulation-based search.

Lecture 10 — Approximate Value/Policy Iteration: AVI performance bound ($\|q^* - q_{\pi_n}\|_\infty \leq \frac{2\gamma}{(1-\gamma)^2} \max_k \varepsilon_k$), API performance bound, FQI, TD(λ) approximation bound (Tsitsiklis & Van Roy, 1997). Note: The modified Bellman operator (Section 2.1) is thematically related but is from Lecture 4, not 10.

Lecture 11 — Off-Policy Multi-Step Methods: Importance sampling ratios $\rho_t = \pi(A_t|S_t)/\mu(A_t|S_t)$, per-decision importance weighting, V-trace / ABTD truncation $\lambda_t = \min(1, 1/\rho_t)$, tree-backup algorithm, deadly triad analysis.

6 Appendix: Update Rule Reference Card

Tabular SARSA (Lecture 6)

$$q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha (R_{t+1} + \gamma q_t(S_{t+1}, A_{t+1}) - q_t(S_t, A_t))$$

On-policy: $A_{t+1} \sim \pi(\cdot|S_{t+1})$.

Tabular Q-Learning (Lecture 6)

$$q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha \left(R_{t+1} + \gamma \max_{a'} q_t(S_{t+1}, a') - q_t(S_t, A_t) \right)$$

Off-policy: behavior can be any exploratory policy.

On-Policy Expected SARSA — Linear FA (Lecture 6)

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha \left(R_{t+1} + \gamma \sum_{a'} \pi(a'|S_{t+1}) q_{\mathbf{w}_t}(S_{t+1}, a') - q_{\mathbf{w}_t}(S_t, A_t) \right) \nabla_{\mathbf{w}} q_{\mathbf{w}_t}(S_t, A_t)$$

With linear $q_{\mathbf{w}}(s, a) = \mathbf{w}^\top \mathbf{x}(s, a)$: $\nabla_{\mathbf{w}} q_{\mathbf{w}} = \mathbf{x}(s, a)$.

Off-Policy Expected SARSA — Linear FA (Lecture 6)

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha \left(R_{t+1} + \gamma \sum_{a'} \pi_{\text{target}}(a'|S_{t+1}) q_{\mathbf{w}_t}(S_{t+1}, a') - q_{\mathbf{w}_t}(S_t, A_t) \right) \mathbf{x}(S_t, A_t)$$

Target policy π_{target} need not equal the behavior policy.

Synchronous Value Iteration — Tabular (Lecture 3)

State values (summation notation):

$$v_{k+1}(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_k(s') \right], \quad \forall s$$

State-action values (expectation notation):

$$q_{k+1}(s, a) = \mathbb{E} \left[R_{t+1} + \gamma \max_{a'} q_k(S_{t+1}, a') \mid S_t = s, A_t = a \right], \quad \forall s, a$$

State-action values (summation notation):

$$q_{k+1}(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \max_{a'} q_k(s', a'), \quad \forall s, a$$

Bellman Equation for v_π (Summation — Lecture 3)

$$v_\pi(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_\pi(s') \right]$$

Bellman Optimality Equation for q^* (Continuous/Integral — Lecture 4)

$$q^*(s, a) = r(s, a) + \gamma \int_{s'} p(s'|s, a) \max_{a'} q^*(s', a') ds'$$

TD(0) Update — Tabular and Linear FA (Lecture 5)

Tabular:

$$v_{t+1}(S_t) = v_t(S_t) + \alpha (R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t))$$

Linear FA ($v_{\mathbf{w}}(s) = \mathbf{w}^\top \mathbf{x}(s)$):

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha (R_{t+1} + \gamma \mathbf{w}_t^\top \mathbf{x}_{t+1} - \mathbf{w}_t^\top \mathbf{x}_t) \mathbf{x}_t$$

TD(λ) Forward-View Target (Lecture 5)

Recursive:

$$G_t^\lambda = R_{t+1} + \gamma \left[(1 - \lambda) v(S_{t+1}) + \lambda G_{t+1}^\lambda \right]$$

Explicit summation:

$$G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \left[\sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n v(S_{t+n}) \right]$$

Policy Gradient Update — REINFORCE (Lecture 9)

Without baseline:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha G_t \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(A_t | S_t)$$

With baseline $b(S_t)$:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha (G_t - b(S_t)) \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(A_t | S_t)$$

Common choice: $b(S_t) = v_{\mathbf{w}}(S_t)$.

Average-Reward Policy Objective (Lecture 9)

$$J(\boldsymbol{\theta}) = \mathbb{E}_{\pi_{\boldsymbol{\theta}}}[R_{t+1}] = \sum_s d_{\pi}(s) \sum_a \pi_{\boldsymbol{\theta}}(a|s) r(s, a)$$

7 Appendix: Key Proof Skeletons

1. T^π is a γ -contraction

Goal: $\|T^\pi v_1 - T^\pi v_2\|_\infty \leq \gamma \|v_1 - v_2\|_\infty$.

Key steps:

1. Write out $(T^\pi v_1)(s) - (T^\pi v_2)(s) = \gamma \sum_a \pi(a|s) \sum_{s'} p(s'|s, a)(v_1(s') - v_2(s'))$.
2. Take absolute value and bound by $\gamma \|v_1 - v_2\|_\infty$ using $\sum_a \pi(a|s) = 1$ and $\sum_{s'} p(s'|s, a) = 1$.
3. Take max over s .

2. T^* is a γ -contraction

Key step: $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$.

Apply to $f(a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a)v_1(s')$ and similarly for g . The $r(s, a)$ terms cancel.

3. Policy Improvement Theorem

Goal: If $q_\pi(s, \pi'(s)) \geq v_\pi(s)$ for all s , then $v_{\pi'}(s) \geq v_\pi(s)$.

Key steps:

1. Start: $v_\pi(s) \leq q_\pi(s, \pi'(s)) = \mathbb{E}_{\pi'}[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s]$.
2. Apply greedy inequality to $v_\pi(S_{t+1})$: $v_\pi(S_{t+1}) \leq q_\pi(S_{t+1}, \pi'(S_{t+1}))$.
3. Iterate to get $v_\pi(s) \leq \mathbb{E}_{\pi'}[G_t | S_t = s] = v_{\pi'}(s)$.

4. Score Function Trick Derivation

Key identity: $\nabla_\theta \pi_\theta(a|s) = \pi_\theta(a|s) \nabla_\theta \log \pi_\theta(a|s)$.

Therefore: $\nabla_\theta \sum_a \pi_\theta(a|s) f(a) = \sum_a \pi_\theta(a|s) f(a) \nabla_\theta \log \pi_\theta(a|s) = \mathbb{E}_\pi[f(A) \nabla_\theta \log \pi_\theta(A|S)]$.

5. Policy Gradient Theorem Proof (Sketch)

Key steps:

1. Start from $J(\theta) = \mathbb{E}_{S_0}[v_\pi(S_0)]$.
2. Expand $\nabla_\theta v_\pi(s) = \nabla_\theta \sum_a \pi_\theta(a|s) q_\pi(s, a)$.
3. Product rule: $= \sum_a [\nabla_\theta \pi_\theta(a|s)] q_\pi(s, a) + \sum_a \pi_\theta(a|s) \nabla_\theta q_\pi(s, a)$.
4. First term gives the score function. Second term: expand q_π using Bellman, recurse.
5. Collect all terms across the trajectory: sum over t with weighting by γ^t and state visitation.

6. Baseline Has Zero Expected Gradient

$$\mathbb{E}_\pi[b(S_t) \nabla_\theta \log \pi_\theta(A_t | S_t)] = \mathbb{E}_S[b(S_t) \underbrace{\sum_a \nabla_\theta \pi_\theta(a | S_t)}_{=\nabla_\theta 1=0}] = 0.$$

7. UCB + Q-learning Convergence

1. UCB ensures every action in each state is tried infinitely often (exploration bonus $\rightarrow \infty$ for untried actions).
2. Ergodicity ensures all states visited infinitely often.
3. Therefore all (s, a) pairs sampled infinitely often.
4. Q-learning convergence theorem applies (stochastic approximation with Robbins-Monro conditions).

8. Modified Bellman Operator Analysis Template

1. **Contraction:** Same proof as T^* —the max is outside the expectation but the key inequality $|\max f - \max g| \leq \max |f - g|$ still applies.
2. **Unique fixed point:** Banach FPT applies since T is a γ -contraction on a complete space.
3. **Differs from q^* :** By Jensen's inequality, $\max_b \mathbb{E}[q(S', b)] \leq \mathbb{E}[\max_b q(S', b)]$, so the fixed point $\hat{q}_T \leq q^*$ in general.
4. **Deterministic case:** The expectation is a point evaluation, so $\max \mathbb{E}[\cdot] = \mathbb{E}[\max(\cdot)]$ and $\hat{q}_T = q^*$.
5. **Counterexample:** Any MDP with stochastic transitions where two actions lead to different states with different values suffices.

8 Appendix: Definitions & Theorems Summary (Lectures 1–9)

All exam-relevant definitions and theorem statements, lecture by lecture. No proofs. Lecture 8 (Planning/Dyna/MCTS) is omitted entirely—never examined.

Lecture 1 — Introduction to RL

Def. (MDP). A Markov Decision Process is a tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$: state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $p(s'|s, a)$, expected reward $r(s, a)$, discount $\gamma \in [0, 1]$.

Def. (Markov Property). $p(r, s' | S_t, A_t) = p(r, s' | \mathcal{H}_t, A_t)$; the state is a sufficient statistic of the history.

Def. (Return). $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$. Recursive form: $G_t = R_{t+1} + \gamma G_{t+1}$.

Def. (Policy). Deterministic: $a = \pi(s)$. Stochastic: $\pi(a|s) = \Pr(A_t = a | S_t = s)$.

Def. (State-Value Function). $v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s]$.

Def. (Action-Value Function). $q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A_t = a]$.

Def. (Optimal Value Functions). $v^*(s) = \max_\pi v_\pi(s)$, $q^*(s, a) = \max_\pi q_\pi(s, a)$.

Thm. (Bellman Equation for v_π). $v_\pi(s) = \mathbb{E}_\pi[R_{t+1} + \gamma v_\pi(S_{t+1}) | S_t = s]$.

Thm. (Bellman Optimality Equation). $v^*(s) = \max_a \mathbb{E}[R_{t+1} + \gamma v^*(S_{t+1}) | S_t = s, A_t = a]$.

Lecture 2 — Bandits

Def. (Multi-Armed Bandit). Tuple $(\mathcal{A}, \{\mathcal{R}_a\}_{a \in \mathcal{A}})$: finite actions, unknown reward distributions. $q(a) = \mathbb{E}[R_t | A_t = a]$.

Def. (Action Regret / Gap). $\Delta_a = v^* - q(a)$, where $v^* = \max_a q(a)$.

Def. (Total Regret). $L_t = \sum_{n=1}^t \Delta_{A_n} = \sum_a N_t(a) \Delta_a$.

Prop. (Incremental Update). $Q_{t+1}(A_t) = Q_t(A_t) + \alpha_t(R_{t+1} - Q_t(A_t))$, with $\alpha_t = 1/N_t(A_t)$ for sample mean.

Thm. (Hoeffding's Inequality). For i.i.d. $X_i \in [0, 1]$ with mean μ : $\Pr(\bar{X}_n + u \leq \mu) \leq e^{-2nu^2}$.

Def. (UCB). $A_t = \arg \max_a [Q_t(a) + c\sqrt{\frac{\log t}{N_t(a)}}]$.

Thm. (UCB Regret). $\mathbb{E}[L_t] \leq 8 \sum_{a: \Delta_a > 0} \frac{\log t}{\Delta_a} + O(\sum_a \Delta_a)$ (logarithmic—order-optimal).

Thm. (Lai-Robbins Lower Bound). $\liminf_{t \rightarrow \infty} \frac{\mathbb{E}[L_t]}{\log t} \geq \sum_{a: \Delta_a > 0} \frac{\Delta_a}{\text{KL}(\mathcal{R}_a \| \mathcal{R}_{a^*})}$.

Def. (Thompson Sampling). Sample $\tilde{Q}(a) \sim p(q(a) | \text{data})$ for each a ; select $A_t = \arg \max_a \tilde{Q}(a)$. With Bernoulli rewards and Beta(1, 1) prior: posterior after k successes, ℓ failures is Beta(1 + k , 1 + ℓ).

Thm. (Policy Gradient for Bandits / Score Function Trick). $\nabla_\theta \mathbb{E}_{\pi_\theta}[R_t] = \mathbb{E}_{\pi_\theta}[R_t \nabla_\theta \log \pi_\theta(A_t)]$.

Def. (Softmax Policy). $\pi_\theta(a) = \frac{e^{H(a)}}{\sum_b e^{H(b)}}$. Score: $\frac{\partial \log \pi(A_t)}{\partial H(a)} = \mathcal{I}(a = A_t) - \pi(a)$.

Lecture 3 — MDPs and Dynamic Programming

Thm. (Bellman Expectation Equations).

$$v_\pi(s) = \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_\pi(s') \right]$$

$$q_\pi(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \sum_{a'} \pi(a'|s') q_\pi(s', a')$$

Thm. (Bellman Optimality Equations).

$$v^*(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v^*(s') \right]$$

$$q^*(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \max_{a'} q^*(s', a')$$

Thm. (Existence of Optimal Policies). There exists π^* with $v_{\pi^*}(s) = v^*(s)$ and $q_{\pi^*}(s, a) = q^*(s, a)$ for all s, a . A deterministic optimal policy always exists.

Thm. (Policy Improvement). If $q_\pi(s, \pi'(s)) \geq v_\pi(s)$ for all s , then $v_{\pi'}(s) \geq v_\pi(s)$ for all s . Equality iff π is already optimal.

Thm. (Direct Solution). $\mathbf{v}_\pi = (\mathbf{I} - \gamma \mathbf{P}^\pi)^{-1} \mathbf{r}^\pi$. Cost: $O(|\mathcal{S}|^3)$.

Def. (Value Iteration). $v_{k+1}(s) = \max_a [r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_k(s')]$. Converges to v^* .

Def. (Policy Iteration). Alternate: evaluate v_π exactly, then improve $\pi'(s) = \arg \max_a q_\pi(s, a)$. Converges to π^* .

Lecture 4 — Operators and Contraction Mappings

Def. (Contraction Mapping). $\mathcal{T} : \mathcal{X} \rightarrow \mathcal{X}$ is an α -contraction if $\|\mathcal{T}x_1 - \mathcal{T}x_2\| \leq \alpha \|x_1 - x_2\|$ for all x_1, x_2 , $\alpha \in [0, 1)$.

Thm. (Banach Fixed Point Theorem). If $\mathcal{X}$ is a complete normed vector space and $\mathcal{T}$ is a γ -contraction, then: (i) $\mathcal{T}$ has a unique fixed point x^* ; (ii) $\mathcal{T}^n x_0 \rightarrow x^*$ for any x_0 ; (iii) $\|\mathcal{T}^n x_0 - x^*\| \leq \gamma^n \|x_0 - x^*\|$.

Def. (Bellman Expectation Operator T^π). $(T^\pi f)(s) = \sum_a \pi(a|s) [r(s, a) + \gamma \sum_{s'} p(s'|s, a) f(s')]$. Fixed point: v_π .

Def. (Bellman Optimality Operator T^*). $(T^* f)(s) = \max_a [r(s, a) + \gamma \sum_{s'} p(s'|s, a) f(s')]$. Fixed point: v^* .

Def. (T^* for Q -values). $(T_Q^* f)(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \max_{a'} f(s', a')$. Fixed point: q^* .

Prop. T^π and T^* are γ -contractions w.r.t. $\|\cdot\|_\infty$. Both are monotonic.

Thm. (Value of Greedy Policy). $\|q^* - q^\pi\|_\infty \leq \frac{2\gamma}{1-\gamma} \|q^* - q\|_\infty$ for π greedy w.r.t. q .

Lecture 5 — Model-Free Prediction (MC and TD)

Def. (MC Policy Evaluation). $v(S_t) \leftarrow v(S_t) + \alpha(G_t - v(S_t))$. Target G_t is the full return. Unbiased, high variance.

Def. (TD Target & TD Error). Target: $R_{t+1} + \gamma v_t(S_{t+1})$. Error: $\delta_t = R_{t+1} + \gamma v_t(S_{t+1}) - v_t(S_t)$.

Def. (TD(0) Update — Tabular). $v_{t+1}(S_t) = v_t(S_t) + \alpha \delta_t$. Biased target (if $v_t \neq v_\pi$), lower variance than MC.

Def. (n -Step Return). $G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n v(S_{t+n})$. $n=1$: TD(0); $n \rightarrow \infty$: MC.

Def. (λ -Return). $G_t^\lambda = (1 - \lambda) \sum_{n=1}^{\infty} \lambda^{n-1} G_t^{(n)}$. Recursive: $G_t^\lambda = R_{t+1} + \gamma[(1 - \lambda)v(S_{t+1}) + \lambda G_{t+1}^\lambda]$. $\lambda=0$: TD(0); $\lambda=1$: MC.

Thm. (Tabular Convergence). Both MC and TD(0) converge to v_π given $\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$, and all states visited infinitely often.

Thm. (Batch MC vs TD). Batch MC minimizes MSE of observed returns. Batch TD converges to the value of the maximum-likelihood MDP.

Lemma (Telescoping). $G_t - v(S_t) = \sum_{k=t}^{T-1} \gamma^{k-t} \delta_k$.

Def. (Eligibility Trace). $\mathbf{e}_t = \gamma \lambda \mathbf{e}_{t-1} + \mathbf{x}_t$, $\mathbf{e}_{-1} = \mathbf{0}$. Update: $\Delta \mathbf{w} = \alpha \delta_t \mathbf{e}_t$.

Def. (Linear Value FA). $v_{\mathbf{w}}(s) = \mathbf{w}^\top \mathbf{x}(s)$. SGD update: $\Delta \mathbf{w} = \alpha(v_\pi(S_t) - v_{\mathbf{w}}(S_t)) \mathbf{x}_t$.

Lecture 6 — Model-Free Control

Def. (GPI). Generalised Policy Iteration: alternate policy evaluation (estimate q_π) and policy improvement (greedy or ε -greedy w.r.t. q).

Def. (ε -Greedy Policy). $\pi_\varepsilon(a|s) = \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}| & a = \arg \max_{a'} q(s, a') \\ \varepsilon/|\mathcal{A}| & \text{otherwise} \end{cases}$

Def. (SARSA). $q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha(R_{t+1} + \gamma q_t(S_{t+1}, A_{t+1}) - q_t(S_t, A_t))$. On-policy.

Def. (Q-Learning). $q_{t+1}(S_t, A_t) = q_t(S_t, A_t) + \alpha(R_{t+1} + \gamma \max_{a'} q_t(S_{t+1}, a') - q_t(S_t, A_t))$. Off-policy.

Thm. (Q-Learning Convergence). Tabular Q-learning converges to q^* given: (i) all (s, a) visited infinitely often; (ii) $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$.

Def. (Expected SARSA). Target: $R_{t+1} + \gamma \sum_{a'} \pi(a'|S_{t+1}) q(S_{t+1}, a')$. On-policy or off-policy depending on π .

Def. (Double Q-Learning). Two independent estimates q^A, q^B . Eliminates maximization bias of standard Q-learning.

Key Distinction. SARSA evaluates the actual behavior policy (risk-sensitive); Q-learning targets q^* (risk-neutral). Both execute ε -greedy; the difference is the update target.

Lecture 7 — Function Approximation

Def. (Value Error). $\overline{VE}(\mathbf{w}) = \mathbb{E}_{S \sim d_\pi} [(v_\pi(S) - v_{\mathbf{w}}(S))^2]$, where d_π is the on-policy state distribution.

Def. (Linear TD(0) Update). $\mathbf{w}_{t+1} = \mathbf{w}_t + \alpha(R_{t+1} + \gamma \mathbf{w}_t^\top \mathbf{x}_{t+1} - \mathbf{w}_t^\top \mathbf{x}_t) \mathbf{x}_t$.

Thm. (MC Fixed Point). $\mathbf{w}_{MC} = \arg \min_{\mathbf{w}} \overline{VE}(\mathbf{w}) = \mathbb{E}[\mathbf{x}_t \mathbf{x}_t^\top]^{-1} \mathbb{E}[G_t \mathbf{x}_t]$. Projection onto feature space.

Thm. (TD Fixed Point). $\mathbf{w}_{TD} = \mathbb{E}[\mathbf{x}_t (\mathbf{x}_t - \gamma \mathbf{x}_{t+1})^\top]^{-1} \mathbb{E}[R_{t+1} \mathbf{x}_t] = \mathbf{A}^{-1} \mathbf{b}$.

Thm. (TD Value Error Bound). $\overline{VE}(\mathbf{w}_{TD}) \leq \frac{1}{1-\gamma} \min_{\mathbf{w}} \overline{VE}(\mathbf{w})$.

Def. (Deadly Triad). Bootstrapping + function approximation + off-policy learning can cause divergence.

Def. (LSTD). $\mathbf{w}_{TD} = \hat{\mathbf{A}}^{-1} \hat{\mathbf{b}}$, with $\hat{\mathbf{A}} = \sum_t \mathbf{x}_t (\mathbf{x}_t - \gamma \mathbf{x}_{t+1})^\top$, $\hat{\mathbf{b}} = \sum_t R_{t+1} \mathbf{x}_t$. Closed-form, no step size.

Def. (Experience Replay). Store transitions $(S_t, A_t, R_{t+1}, S_{t+1})$ in buffer; sample uniformly to update. Converges to max-likelihood MDP solution if buffer is representative.

Def. (DQN). Q-learning with neural network FA, experience replay buffer, and a periodically-updated target network q_{θ^-} .

Lecture 9 — Policy Gradient Methods

Def. (Average-Reward Objective). $J(\theta) = \mathbb{E}_{\pi_\theta} [R_{t+1}] = \sum_s d_\pi(s) \sum_a \pi_\theta(a|s) r(s, a)$.

Lemma (Score Function Trick). $\nabla_\theta \mathbb{E}_{\pi_\theta} [f(S, A)] = \mathbb{E}_{\pi_\theta} [f(S, A) \nabla_\theta \log \pi_\theta(A|S)]$.

Thm. (Policy Gradient — Episodic). $\nabla_\theta J(\theta) = \mathbb{E}_\pi \left[\sum_{t=0}^T \gamma^t q_\pi(S_t, A_t) \nabla_\theta \log \pi_\theta(A_t|S_t) \mid S_0 \sim d_0 \right]$.

Thm. (Policy Gradient — Average Reward). $\nabla_\theta J(\theta) = \mathbb{E}_\pi [q_\pi(S_t, A_t) \nabla_\theta \log \pi_\theta(A_t|S_t)]$, with differential q_π .

Prop. (Baseline Zero-Mean). $\mathbb{E}_\pi [b(S_t) \nabla_\theta \log \pi_\theta(A_t|S_t)] = 0$ for any $b(s)$ independent of A_t .

Def. (REINFORCE). $\theta_{t+1} = \theta_t + \alpha G_t \nabla_\theta \log \pi_\theta(A_t|S_t)$. With baseline: replace G_t by $G_t - b(S_t)$.

Def. (Actor-Critic). Actor: policy π_θ . Critic: learned $v_{\mathbf{w}}(s)$. Use TD error $\delta_t = R_{t+1} + \gamma v_{\mathbf{w}}(S_{t+1}) - v_{\mathbf{w}}(S_t)$ in place of G_t .

Gaussian Score Functions. $\nabla_\mu \log \pi(a|s) = \frac{a-\mu}{\sigma^2}$, $\nabla_\sigma \log \pi(a|s) = \frac{(a-\mu)^2 - \sigma^2}{\sigma^3}$.

Softmax Score Function. $\frac{\partial \log \pi(A_t)}{\partial H(a)} = \mathcal{I}(a = A_t) - \pi(a)$.

Def. (DPG). $\nabla_\theta J(\theta) = \mathbb{E}_{S \sim d_\mu} [\nabla_\theta \mu_\theta(S) \nabla_a q_\mu(S, a)|_{a=\mu_\theta(S)}]$.